The Internal Logic of Buddhism Is Verifiably Correct

And I've proven it. Conventionally speaking.



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$$\neg\Box(v(P) = \mathbf{t}) \wedge \neg\Box(v(P) = \mathbf{f}) \wedge \neg\Box(v(P) = b) \wedge \neg\Box(v(P) = n)$$

Hello. Welcome to *Dissolved Distinctions*.

Today is the official publishing date of my paper "The Modal Catuṣkoṭi: Formalizing Emptiness in Classical Modal Logic" to the journal *Comparative Philosophy*, peer-reviewed by scholars with relevant expertise in Buddhist philosophy and formal logic.

I've been waiting until now to continue posting anything on this blog, particularly as it has been quite difficult to come up with anything other than this to talk about. Hence the provocative title, which I stand by in full.

My paper, on its face, supplies an internally coherent formula that untangles the Western understanding of Nagarjuna, a thinker 1800 years ago whose thought supplies the core of Tibetan and Zen Buddhist lineages around the world. It establishes Mahayana Buddhist philosophy

as compatible within the same general class of logic as Aristotle by modalizing the "four corners" that the great Buddhist presents, so that it is demonstrated that he rejects those four positions as being necessary fixations. It also does more than that, less explicitly.

My paper establishes that the internal logic of philosophical Mahayana Buddhism (itself having always claimed to be an aspect of the Buddha's teachings) is correct.

By this, I don't mean every cosmological or doctrinal assertion made by every sub-tradition or school. What is instead being spoken of is its structural operation.

These claims will sound extravagant if they are heard as metaphysical ones about the necessary reality of God or gods, souls, consciousness, karma, rebirth and the afterlife, or other such topics. They are not. I'm not claiming that Buddhism supplies a foundation to supplant all others, or that there's some ultimate substance hiding underneath other views. Instead, what's being claimed is precise in target while expansive in scope: once we enter discourse, make claims, and ask for rational warrant, the conditions of that activity force a single conclusion. No claim can secure itself as necessarily grounded. [1]

That is what I mean when I say the internal logic of philosophical Mahayana Buddhism is correct. Correct, conventionally speaking: not as a final metaphysical thesis, but as the formal refutation of final metaphysical theses. Non-necessity and reflexive contingency supply the formal countermodel to foundationalist thought.

Let me explain, in language that isn't Buddhist or mystical. What I offer is nothing fundamentally complex. However, it has been machine-checked without caveats to the most rigorous level I'm aware of within logic and mathematics. Its wider philosophical force rests on two very thin bridges, which I'll proceed to cover: a foundation that does foundational work claims non-contingent necessity; and an admissible

counterpossibility defeats that necessity. Once we accept these quite basic bridges and definitions, the derivation is true in the boring and exact way that 2+2=4 is – once the structure is visible, the result follows. No appeal to mystical authority required.

So, What Is It?

I will present the technical version, then the technical translation of the technical version, then the equivalent in general English.

The technical version:

¬□(v(P) = t) ∧ ¬□(v(P) = f) ∧ ¬□(v(P) = b) ∧ ¬□(v(P) = n)

[¬ represents negation, □ represents necessity, v stands for truth value, P is any arbitrary proposition, ∧ means 'and', and t/f/b/n are what you'd expect: true, false, both, neither.]

The technical translation:

No proposition has a truth value assigned to it necessarily.

The general English equivalent:

No developed speech or thought can be secured as *fundamentally* the case. Including what I'm saying now.

Wait – How Does That Work?

A lot of people might think I just said something contradictory. When people think this, they're usually thinking of something that has a name in analytic philosophy: the Liar's Paradox, like when a person says: "This sentence is false."

But notice a difference. The formula above doesn't deny that things can be true, or false, or true and false (say, as a matter of perspective), or

neither true nor false (in a category error fashion, like asking if blue can fly). The formula instead denies that such determinations are intrinsically, fundamentally, or essentially the case.

Yet the formula also refuses to omit itself from the same analysis. This leads to a peculiar illusory tension in natural language – the logic, the formula, does not recognize this tension, but it appears within speech.

The tension is this: I cannot tell you that this formula is *necessarily* true, for obvious reasons. Or that it's necessarily false. However, structurally, any attempt by speech or thought to escape the formula ends up developing enough content to be subjected to the formula. The logic is verified. Conventionally speaking, it is true.

Wait – How Does This Apply to All Thought and Speech?

> *Master Huaitang said to an assembly,*
>
> *The proposition of being and the proposition of nonbeing are like vines clinging to a tree – you people may nod your heads for the moment, but when the tree falls and the vines wither, you have no strategy to challenge the heavens and no tactics to enter the earth. If a spiritually sharp individual can set an eye here, he will see freely in all ways.*
>
> *[Raising his whisk:] Sunlight fills the eyes – there's not a cloud for myriad miles. If this is under an overturned bowl, how can you blame me?*
>
> - Case 111 of Da Hui's *Treasury of the Eye of True Teaching*, translated by Thomas Cleary

This is for a few simple reasons, and they fall directly out of the bottom of what we observe about the formula. This doesn't require some crude

claim that every passing sensation or flicker of experience is a sentence in the head. The point is structural rather than psychological.

Whatever enters thought or speech as content - whatever can be meant, indicated, remembered, distinguished, defended, denied, or offered as a counterexample - has already entered a field of relations. It is this field that gives it determinacy as anything at all. In fact, while the formula uses “P”, it can trivially swap P for “Pc” – propositional content.

If someone appeals to a form of thought or experience outside such content, that appeal faces a simple fork. If it cannot be indicated in any way, it cannot ground an objection. If it can be indicated, even as “ineffable,” “pre-conceptual,” “embodied,” or “beyond propositions,” then it has entered propositional content precisely enough for the formula to apply.

These lead us to our conclusions. They are philosophical conclusions, and yet also extremely straightforward ones that are difficult to refute as soon as we stop assuming a secret metaphysics lurking behind each statement; working through the formula helps establish how clear they are. (We’ll walk through why.)

For one: All speech is propositional, and as such oriented around propositional content. Speech has referents and seeks to convey semantic meaning. Speech without meaning is not considered speech, or certainly not to any level that does productive work with the term.

And for the other: The same structure reaches thought itself.

To form a thought is to establish subjects, objects, and points of reference. In so doing, you form something coherent, and to form something coherent you must take what you’re talking about to necessarily be the case as a form of operational fixity, if only to be coherent at all. For instance, even when thinking of unicorns with glass eyes that breathe cold fire, you still have to treat unicorns, glass eyes,

and cold fire as stable concepts that necessarily carry meaning.[2] Some people say they don't have an inner voice or monologue – but work is still being done cognitively to form the equivalent of "I am looking at a digital screen," in which your existence and your act of looking at a digital screen are not in dispute.

Propositions, in other words, are tied to the hip with *semanticity* – with the expression of meaning. Here is what I mean:

Imagine a word like "dog." Try to find what *is* underneath everything it connects to. Strip away that it's an animal, that it barks, that it's not a cat. Essentially, strip every relation. The trouble is that those relations were what told you it was a dog in the first place; pull them all and there's nothing left that was ever a dog to begin with. You can't ever get to a bare "dog" sitting under the meaning. It's the same the other way: there's no scrap of meaning floating free with no claim it belongs to, because a meaning that says nothing about anything isn't yet meaning anything.

So we never actually run into something standing outside all of this. This is not because reality is built that way, but because the moment you try to point at the bare thing, the pointing has already wrapped it back up in sense.

The skeptic will protest: "What about thoughts that can't be articulated? Surely then it's not propositional." They might even bring up Thomas Nagel's famous example, asking what it must be like to be a bat, to suggest that there must be qualia, something that establishes 'what it's like'-ness.

But what is an unarticulated thought? Are you able to describe it without engaging in mental or spoken articulation of what an unarticulated thought even is? Please feel free to try, if you wish. If you can't describe it without articulation, what reason do you have to say that unarticulated thought necessarily exists? In fact, it now looks like

to do so is to make a claim imposing a *necessarily assigned truth value* that distinguishes what is and isn't articulated, which the formula rejects.

If we don't have a reason to say unarticulated thought necessarily exists, and if we have no proof of direct access since proof would be articulable, why posit it as a necessary presumption at all?

That's not at all to say that unarticulated thought or qualia strictly and ultimately can't exist. "Can't" would then be a necessity claim. (This is why 'mystical' speech in Buddhism or in other traditions making similar moves often sounds so paradoxical.) It's to say that, given the above unpacking on propositional content, things aren't found as necessary existents by the same modes of thought by which we posit necessary existents - we never get to that bare dog. Thought has no structural ability to escape this mingling of propositional content yet doesn't write things off merely by fiat. There is structural interdependence.

So What Does This Mean?

> *"Whatsoever depends on conditions,*
> *That is devoid of intrinsic existence."*
> *What excellent instruction can there be*
> *More amazing than this proclamation?*
>
> *By grasping at it the childish*
> *Strengthen bondage to extreme views;*
> *For the wise this very fact is the doorway*
> *To cut free from the net of elaborations.*
>
> Je Tsongkhapa, *In Praise of Dependent Origination*, verses 5-6, translated by Geshe Thupten Jinpa

All metaphysically fixed foundations, and as such all foundationalisms in any absolutist sense, are structurally forfeit under their own standards due to being defined around necessary truth value claims operating asymmetrically. Necessity cannot establish its own jurisdiction.

This sounds like a very bold claim, but it merely follows from everything said prior. The consequences make clear that thought is structurally interdependent on propositional content, and propositional content does not hold a necessarily assigned truth value. All one needs to refute "necessarily-P" is "possibly-not-P," at each self-contained step, all the way down.

To be very clear on this: this is NOT saying no truths "actually exist", or that people simply can't make posits that *may* be true. To say that would be a foundationalist position. The formula itself is specifically about necessarily assigned truth-values to P, after all, not about P itself.

The way it works is this: Necessity isn't refuted proposition by proposition. It's consistently disqualified as self-grounding logical authority, because every attempted justification of necessity occurs within propositional content, which is subject to the formula.

A committed opponent might say that necessarily-P holds in this actual world, without making claims as to whether "this is the actual world" is itself necessary. It should be clear that this is just relocating the formula's mechanism one step up an infinite ladder: the opponent has no justificatory strength to justify "necessarily-P holds in this actual world" when the formula makes clear that no such proposition is necessarily true. For every bare or brute-force posit, this formula forces those posits to show their work in properly deriving metaphysical necessity, which contradicts what "bare" and "brute-force" posits are meant to do.

For instance, an essentialist might cheerfully say that even if their positions are contingent, the actuality and truth of what they're

speaking about remains the same. One simply then asks: True *in what sense*, and *established how*? If the essentialist ever had a bottom to directly touch, at any structural point, to *prove* their thesis, a proposition would be logically terminable and the Lean proof would not be valid. The formula, however, makes clear that ultimate, robust terminability does not exist within the logical structures that give rise to such questions and answers in the first place.

The only way out isn't a way out at all, but merely a recognition that the structure and meta-structure (and meta-meta-structure, and so on, and so on) is a cognitive trap when settled upon without self-aware revisability.

As a result, this is not a mere opinion. It is a conventional truth, which is the strongest one we have. All one can do to verify this themselves is to watch what happens when another attempts to justify their disagreement.

The Buddhist position has always been that the ultimate grounding that foundationalist thinking uses to terminate further productive inquiry – including all contemporary dreams that foundations can prove souls, gods, moral truths, an origin of all things, life before conception, a difference between 'us' and 'them,' a fundamental tying of sex and gender, of East versus West – is but an illusion: it is not structurally available but attains the impression of it.

One might say that this applies symmetrically, that this means there's no fundamental *untying* of sex and gender, of East and West, etc., or an ability to fundamentally disprove souls and gods and moral truths either. That is correct.

In practice, however, social and political positions stand in different structural relations to essence-claims. Any ideology that acquires most or all of its merit by insisting on foundational structures that define what reality must be is incorrect beyond mere opinion. No hierarchy

gains authority by calling itself natural, essential, or metaphysically unavoidable. When something or someone does so, they fail to be coherent.

You Say It's Machine-Verified. But I'm Not Convinced.

> *3.2.3.3.2.2.1*
>
> *If one has the eyes of the authentic view,*
> *Scriptural learning, experience, and realization blaze up.*
> *By virtue of seeing emptiness,*
> *Confidence in the* ***infallible relativity*** *of cause and effect*
> *Will increase, and emotional disturbance will lessen.*
>
> *3.2.3.3.2.2.2*
>
> *With the samādhi that abides one-pointedly*
> ***In the state of certainty induced by analysis,***
> ***The ultimate meaning is seen by nonseeing.***
>
> *3.2.3.3.2.2.3*
>
> *One does not succumb to any particular object of seeing*
> *And of course does not apprehend anything.*
> *Like a mute's taste of molasses,*
> *Confidence grows in a yogi who cultivates it,*
> ***But it cannot be produced by analysis alone****.*
>
> Jamgön Ju Mipham Gyatso, *Beacon of Certainty*, translated by John W. Pettit (emphasis added)

It's natural to doubt such bold claims. First, I'll walk through a general sketch of the situation and show how the machine verification works.

The formula of the paper is a formula of *modal logic*. Modality in logic involves anything pertaining to necessity and possibility. There are

multiple systems of modal logic, because there are different ideas about what it even means for things to be “necessary” and “possible.”

Systems of modal logic discuss *possible worlds* – ways to talk about internally coherent positions that *could* be true in a world that isn’t the one we currently experience. For instance, we can say there’s a possible world in which Earth was 10 inches further from the Sun than it currently is.

Now, whether we can say this unchallenged is a big problem when discussing possible worlds – many argue that only the current, actual world exists, and possible worlds are (ironically) impossible in actuality. However, it’s clearly *coherent* to think of a universe where the Earth were 10 inches further away from the Sun; otherwise, we couldn’t talk about it. The formula itself shows that these metaphysical arguments are unnecessary; if no thought, and thus no propositional object of thought, has a necessary truth value, this is the same as saying that contingency rules experiential reality, within and around any arbitrary division between possible and actual worlds.

The simplest system of modal logic is **K**. All K cares about is this: A formula is necessarily true if it holds in every accessible world. ‘Accessible’ here is just the term modal logic uses for which worlds count from where you’re standing, and different systems of modal logic have different ideas about what ‘counting’ amounts to.

In system **T**, necessity is reflexive. K permits a formula to be true in other accessible worlds but not necessarily this one; T insists that a formula that is true in all accessible worlds is also true in this one as a ‘seeing-itself’ sort of trait.

In system **S4**, necessity is *necessarily necessary*, or transitive. This is often of epistemological importance – it allows you to say that if someone knows something, they know that they know it.

System **S5** is the strictest by far of this main list, because in S5 *all* possible worlds are also accessible – so if a formula is necessarily or possibly true, it's so absolutely and uniformly in every such world. Accessibility becomes an equivalence relation.

Stronger systems exist, but these are the standard, and stronger systems are often seen as *too* strong – things collapse into just being straightforwardly true or false everywhere, making things too trivial. There are also weaker systems – non-normal logics, neighborhood semantics – that are even more permissive than K when it comes to how things operate as necessary or possible.

If you take our formula, something pretty interesting happens.

The formula itself checks out conceptually under every single relevant system of modal logic that I'm aware of. I verified it from K-S5, but since weaker systems than K are generally even more permissive, we don't need to mechanically verify to suspect as matter of principle that it holds there. Stronger systems than S5 have the above stated issues.

How does the formula have such broad reach? Because everything I've discussed is how systems constrain necessity and possibility – but none of them explicitly touch how systems constrain *non-necessity*. The formula slips through the gaps, as to be expected from contingency.

But then it gets even more interesting.

When you apply the formula to itself, which is where the magic happens – when the formula self-iterates and denies its own status as a fundamental existent – it works in every system of modal logic weaker than S5, and fails in S5. This is because in S5, the bare formula would be true absolutely and necessarily in all accessible worlds, leaving no room to turn back on itself – every further chain of worlds built would satisfy the formula, but never get to its self-iteration. (This limitation

generally is itself a basic feature of S5 identified since at least the 1960s, if not much earlier.)

Key here is that the failure in S5 isn't something to identify as a problem, as it's the *same structural fact* represented differently in different systems. In weaker systems, the formula's universal applicability is shown through staged iteration: the formula refutes necessity sequentially at every level ad infinitum, because asymmetric accessibility leaves room for each stage. Being *infinitely satisfiable in those systems*, it appears in S5 as the practical consequence – of the formula holding throughout the relevant equivalence class, entailing uniformly what staging in weaker systems show progressively. The universal scope of the operation's success is the same fact in both cases; inexhaustible iteration in weaker systems is uniform truth in stronger systems.[3] The same structural fact is represented in each system's native modal vocabulary.

The foundationalist here might breathe a sigh of relief. "Excellent! I can just say that foundations are real, and they exist in S5!"

That would be false.

Why? Because, dating back to Aristotle, foundations have relied upon axioms like that of the "unmoved mover" to explain metaphysical fundamentals – a thing that moves all things but cannot be moved itself. In other words, foundationalism is *asymmetric*; everything must depend on the foundation without the foundation depending on everything. If there were full symmetry and uniformity, there would be nothing to be dependent upon, because everything would be equivalent.

It's possible to say that foundational grounding is an asymmetric "add-on" to S5, and some metaphysicians do this. But at that point, S5 isn't supplying any foundation, and the further commitments being made by the metaphysician are all necessity-commitments subject to scrutiny by the formula.

In other words: No foundationalist position survives this formula in any system that does the work foundationalism requires. They're refuted in systems weaker than S5, as no necessity claim can *be* necessary when possibility is always structurally available. They're refuted in S5 by S5's structure itself, as they need asymmetric effects from a Euclidean base system. The formula in contrast *does* survive as satisfiable in S5, and as infinitely satisfiable in every other relevant system. This relegates the foundationalist position to at best mere parity, which is all that's needed: mere parity at all relevant levels is a structural disproof of any foundation as a necessary existent.

Even in non-classical systems where the formula has to change to accommodate system rules, foundationalism would still need such systems to support asymmetric grounding-style necessity. But asymmetry is exactly what permits any formula-analogue to refute that necessity in the first place – asymmetry is itself the availability of contingency, of something existing as other than the ground. Foundationalists cannot have the asymmetry their position requires without supplying the conditions of their own refutation. The horse is incapable of ever leaving the barn.[4]

This leads to a remarkable irony about the entire project of foundational thinking. It's often held that S5 is the natural formal home for foundations: a fact there can be true everywhere, accessible from everywhere, and the same across all perspectives. Plantinga famously exploits this with his modal ontological argument. As established, however, as a bare system S5 can't establish asymmetric grounding, and attachments to the system are analyzable by the formula. (Some foundational perspectives might use weaker models, but as shown, our formula renders them impotent.) Philosopher Kit Fine has already noticed this in works like "Essence and Modality" (1994). Fine's familiar objection is that essence cannot be reduced to necessity: a truth may be necessary without belonging to a thing's essence. However, it's quite clear even from Fine's perspective that while necessity may not entail

essence, essence does entail necessity. If nothing can be structurally fixed a necessary truth value assignation, essence existing as necessary isn't derivable from the metalogic. A content-preserving counterpossibility remains sufficient to defeat the essentialist claim.

With all routes exhausted, we see that foundations are structurally untenable under their own systems of operation. They fail to logically refute the satisfiable presence of ordinary contingency on their own levels of analysis, despite being predicated around the elimination of contingency in order to perform the logical work.

That Said, Let's Address Some Possible Objections

> *Dependent origination we declare to be emptiness.*
> *It [emptiness] is a dependent concept; just that is the middle path.*
>
> *There being no dharma whatsoever that is not dependently originated,*
> *it follows that there is also no dharma whatsoever that is non-empty.*
>
> Nagarjuna, *Mūlamadhyamakakārikā* (*Root Verses of the Middle Way*), MMK 24:18-19, translated by Mark Siderits

Objection 1: "Well, you can't mean it's actually like 2+2=4. That's valid in Peano arithmetic, pure proof theory! And isn't it a mistake to claim this is formally proved as valid in that way, when really the formula is just about *satisfiability*?"

Response: Lean, given the standards of Peano arithmetic, verifies the proof as a metatheorem about satisfiability and not as a new metaphysical axiom. This is emphatically not a concession. A foundationalist claims any relevant alternative is impossible; therefore, one admissible alternative is sufficient to defeat the claim. The machine check verifies the recursive model construction. There is nothing else

needed by classical logic's standards, or by any definitional unpacking of what "necessity" and "impossibility" means, to refute the charges at hand.

Objection 2: "Well, maybe I'm more of a pragmatist and just accept foundations as most convenient for discourse. We have to use starting points to perform any work!"

Response: Beyond the issues with saying 'have to' – depending on what we mean, Feyerabend made clear that people have indeed refused various starting points yet still performed work – the result of this is accepting foundations as contingent, which is acceding to the formula. Foundations certainly produce conventional work. What's refuted is the attitude in which foundations are held to not be subject to change, revision, uncertainty, or argumentation. If you tell me that artificial intelligence must be conscious, or must be unconscious, and continually find yourself falling back to other 'musts' to justify yourself, you will continually find that nothing that you *say must be so* can be defended beyond mere assertion, and this isn't a sheer accident or difficulty in expressing yourself.

Objection 3: "Foundations nevertheless remain conventionally necessary for discourse and meaning to work – if we deny this, it produces a performative contradiction. So it's not like we can discard foundations."

Response: This is in fact directly embedded in the formula itself (by being self-applying) and is part of its broader well-formedness as discussed in the responses to Objections 5 and 9. All speech and thought intrudes into metaphysical affirmations in a fashion that is self-illusory once the logic is revealed. This doesn't bar us from using speech or thought pragmatically. What else would there be for us to do?

Discourse functions under presumptions of stability, certainly. Operational dependence at the same time isn't proof of intrinsic ground;

a rule may be indispensable within a practice without being self-grounding across every admissible practice. The formula accepts operational dependence while rejecting the inference to intrinsic ground.

Objection 4: “Well, sure, this formula shows that no proposition is necessarily assigned any truth value given possible worlds. However, it happens to be that this actual world has contingently led to true necessities, so foundational thinking still holds.”

Response: This misunderstands the formula quite directly. Designating one world as actual doesn’t make the necessities asserted there self-grounding, because “this actual world has contingently led to true necessities” is itself a claim subject to the logic. One remains trapped by the propositional bind.

Objection 5: “How can you say all these things so assertively, if the entire theorem is around non-necessity of claims? Isn’t calling this all true a necessity-claim?”

Response: No. Calling something true conventionally is not the same as claiming it is necessarily grounded, and I mean this in the conventional sense as well. Both forms of speech are metaphysically inert, but only conventional truth admits to it.

The claims are correct. The theorem has broader satisfiability-validity within the rigor of dependent type theory. Foundationalisms are structurally forfeit under logical analysis using the structures of ordinary modal contingency. The philosophical justifications fall directly out of the bottom of the theorem when given thought, with only the thinnest and most widely agreed-upon premises when it comes to Nagarjuna’s operation of discourse. Even if those premises turned out to be wrong, such as if in some exceedingly strange scenario Nagarjuna *wasn’t* engaging in refutation of all claims asserting necessary t/f/b/n values (refutation, after all, is his mechanism of dialectic), the formula remains

valid as a theorem. This allows us to make positive statements of truth without claiming metaphysical basis.[5]

Objection 6: “Perhaps propositional content is derivative from a more basic mode of engagement. In other words, maybe phenomenological views like that of Heidegger or others are correct. Perhaps there’s an embodied mode of action that isn’t truth-apt but derives truth-apt content from it.”

Response: It should be noted here that the impact of the formula is not to defeat phenomenology as an available register of dialogue. However, claims beyond mere conventional register face a structural fork: If pre-propositional ground cannot be indicated as a proposition or as propositional content, it cannot ground an objection in any coherent sense. If it can be indicated as one, it is subject to the formula and therefore refuted in any of its pretensions to being necessarily the case.

Objection 7: “Well, okay, so T(P) is satisfiable in S5. But so is its inverse, let’s say ¬T(P). Surely that can’t mean T(P) is true?”

Response: It’s very true that the inverse – essentially the foundationalist’s hopes out of S5 – is also satisfiable. This isn’t a concession and is in fact precisely the point. Here’s why.

As far as weaker systems go, it’s defused as discussed above: ¬T(P) is always capable of being refuted by T(P), and once refuted, it’s not true again. As far as S5 goes, the foundationalist suddenly finds themselves insisting that ¬T(P) can be necessarily true even though it’s shown to simply be one of two possible options to place onto S5 on the same level of analysis. The possibility once again refutes the necessity on every conceivable meta-layer, and you can’t have both engaged at the same time.

In short: any foundationalist that doesn’t give up the ghost on metaphysical foundationalism entirely can’t say “Well, my foundation is

potentially satisfiable in this model, so there!" Then they wouldn't be saying their foundation is **necessary**. The consequences are absurd by the foundationalist's own standards, because if they found mere parity to be an acceptable outcome, they wouldn't be a foundationalist. They'd be acceding directly to relativity and choice. A foundation cannot be merely one satisfiable option among others and still function as a foundation.

Objection 8: "Wait – the Lean proof says that all propositions (phi, in the proof) are stipulated as contingent! Aren't you pre-stipulating this without warrant?"

Response: Is this suggesting that my stipulation of propositions as contingent is itself contingent? That sounds flippant, but this recursiveness is the entire point as detailed above regarding propositional content capture.

This reveals the burden in showing that any proposition we come up with, at all, isn't contingent: it'd have to exist in a space where even thinking of a denial is unthinkable. Even "2+2=5" is thinkable; see **footnote 2** prior for a critique of the Kripkean objection against thinkability and how such objections rely on presuppositions subject to the formula. Moreover, we'd be suggesting something operating in the sense that S5 does wherein as soon as it's brought up, it's true everywhere, symmetrically and uniformly within the admitted modal class. This would have to be so while *also* being equally true in every other modal system in which symmetry isn't forced, *without* eliminating the concept of a ground as impinging on things yet un-impinged on. S5 can't do this. Only this formula can do this in its exhibition across modal systems, and only through its endless satisfiability. What we'd be looking for is exactly what foundationalism can't be.

Objection 9: "What about tautologies? Tautologies are necessarily true. In fact, since it's a Lean-checked meta-theorem, your proof is a tautology, since theorems are tautologies!"

Response: Classical tautologies only remain necessary in a contingent fashion – within fixed classical semantics. The formula doesn't dispute this. What doesn't follow is any further proposition that any single semantics is the uniquely correct and metaphysically necessary framework. "Tautologies, being necessarily true within a given semantics, are thus necessarily true period" is not a tautology and is subject to the formula; after all, classical reasoning is well capable of reasoning classically about non-classical fields even if those fields may deny the validity of certain classical tautologies.

Further, it's difficult to see what metaphysical warrant can be justified from strict tautologies like "This is fundamental or this is not fundamental," or "If it's a person, it has a soul; if it's not a person, it doesn't have a soul". That would require unpacking what a person and a soul is, which is all activity subject to the formula. Theorems and meta-theorems themselves as frame-relative tautologies just elucidate what the frames structurally enforce when in use.

Objection 10: "Sure, this formula is satisfiable in the way you claim. But satisfiable does not mean admissible. Propositions can have satisfiable alternatives but not admissible ones."

Response: Who decides this admissibility? It must be governed by a rationale, so we humbly request to see it. That rationale will be investigated in turn if it rests on further assumptions. This is the formula in operation, precisely as described; the opponent has no rational way to omit themselves from accountability. The denial of admissibility toward a satisfiable result to ensconce ultimate ground may work in a local frame of arbitrarily fixed allowances and disallowances – thus establishing the frame as contingent - but any pretensions to an ultimate fixity are lost to circularity, vicious regress, or bare stipulation. What this means is: So long as a counterassignment remains satisfiable under the shared formal rules, no opponent can prove the corresponding truth-value assignment necessary without either

identifying a formal defect or imposing a further restriction, whose authority is itself local and contestable.

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Someone might ask: Well, what *could* disprove this, if these objections aren't sustained? That's a good question.

Critics must either find a formal error in the encoding or proof, reject one of the bridge principles at seemingly unsurmountable practical cost, or produce a foundation that does foundational work without becoming content, relation, warrant, or necessity. I do not know what that would mean, but I am stating the condition under which it would count. Given the formula is Lean proofed under a quite elementary mathematical function and since mechanical proof checkers like Lean and Coq rely on highly rigorous inductive types, finding a flaw via disproving the proof itself would be a paradigm shift in the history of mathematics. (Moreover, this may also translate as verifiable in Isabelle, whose classical higher-order logic is *more* native to the task than Lean.)

The Lean proof is here:

https://github.com/rollofthedice/modal-catuskoti-lean

There are possibilities at first glance that revolve around rebutting the extremely analytic philosophical bridge claims, but they fall apart on analysis. Someone could try to argue that foundational claims exist outside of modal logic in some capacity, but then they'd have the perplexing situation of positing that as a necessary, and thus modal, claim, all while attempting to abandon a category of logic that was built to justify foundations. Someone could attempt to bracket the scope of what propositions or propositional content touch, but that fails for the reasons discussed prior and is expressly not what the formula does.

This impossible situation for the critic is precisely what you would expect if the formula were capturing something structural about propositional content as a facet of experiential reality. To ask for further proof beyond this would not be coherent by speech's own standards.

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The American Pragmatists once made clear that rejecting a metaphysics doesn't prohibit warranted conclusions of broad scope made with confidence. My paper itself shows how Nagarjuna's "No Thesis" claim is thoroughly coherent without being a metaphysical posit, garnering its refutational strength through that coherence.

The conclusions herein also do not, for obvious reasons, place Buddhism or Mahayana as "better" than other religious traditions. Striking structural parallels to what has been unpacked from this formula can be seen in the apophatics of Meister Eckhart and Mansur al-Hallaj, in the transformations of Zhuangzi, and in the most sophisticated of Advaita Vedanta and Kashmir Shaivist Hindu thought. The formula represents structure, which is not possessed by anyone.

I leave you with a koan very dear to me, from the same compilation of Da Hui cited prior:

> 雲居膺和尚示眾云：老僧二十年前住三峯菴時，有魏府興化長老來問，權借一問以為影草時如何？老僧當時機思遲鈍，道不得，為伊置得箇問頭奇特，不敢辜佗。伊云：想和尚答遮話不得，不如禮拜了退。而今思量，當時不消道箇何必。後因化主到魏府興化，乃借問：山中和尚住三峯菴時，老僧曾問伊話，祇對不得，而今道得也未？化主遂舉前話，興化云：雲居二十年只道得箇何必。興化即不然，爭如道箇不必？三聖云：雲居二十年道得底，猶較佗興化半月程。

妙喜曰：何必不必，綿綿密密，覿面當機。有人續得末後句，許你親見二尊宿。

Master Yunju Ying said to an assembly,

Twenty years ago, when I was living in a hermitage on Three Peaks, Elder Xinghua of Weifu came and asked, "How is it when provisionally using a question for shadow-casting grass?" At that time my mind and thought were slow, and I couldn't give him an answer. Because he had posed the question so remarkably, I dared not offend him. He said, "I think you can't answer this question – it would be better to bow and withdraw." Now as I think of that time, it was hardly worth saying "What's the necessity?"

Later when a fundraiser went to Weifu, Xinghua asked him, "When the teacher on the mountain was living on a hermitage on Three Peaks, I asked him a question he couldn't answer. Has he been able to answer it by now?"

The fundraiser then quoted the foregoing talk. Xinghua said, "In twenty years Yunju has only managed to say 'what's the necessity?' I disagree. How does that compare to saying 'not necessary'?"

Sansheng said, *"What it took Yunju twenty years to say is only comparable to half a month's journey for Xinghua."*

Dahui remarked, *"'What's the necessity?' 'Not necessary' – very tight, very close, meeting the situation face-to-face.*

If anyone can continue the last word, I'll admit you have met the two venerable adepts."

Koan 115, translated by Thomas Cleary.
Chinese pulled from CBeta at https://cbetaonline.dila.edu.tw/zh/X1309_001.

In my next blog post, I'll exhibit how this operation can untie binaries within the most important doctrinal disputes in Buddhist history. I look

forward to seeing you then.

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1 I'll cover what this establishes in relation to Mahayana Buddhist doctrine more precisely in my next blog post.

2 Some might object, bringing up Kripke's argument regarding water and H_2O – a necessary truth *a posteriori* in which thinking of an alternative still makes that alternative impossible. Kripke, a scientific-essentialist, argued that even if we woke up tomorrow and found water was entirely different, he'd still be right in saying that "*If* water is H_2O *then* it's necessarily H_2O", since there's a "microstructure" that is the individuating essence in any case. The formula's structure should make clear that such an assertion, however, cannot hold irrefutable power simply by being presented as an axiom. It is subjected to the formulaic operation and shown to not be necessarily correct as a posit, correctly predicting its unfalsifiability.

3 In this sense, asking "Surely it's a problem the formula can't reiterate in S5?" is formally a category error – S5, bare, is structurally incapable of reiterating upon such necessity results in a productive capacity. If the satisfiability in weaker systems is like being on every infinitely wide and tall story of a tower simultaneously, the satisfiability in S5 is like also being atop the tower's peak, merging with the sky without cognizable beyond. Stronger systems collapsing into triviality are then seen as the point where necessity itself stops doing modal work, and the metaphor vanishes.

4 This is structurally identical to the "fire and fuel" argument by Nāgārjuna in the Mūlamadhyamakakārikā, chapter 10: if fire is the same as the fuel to light it, everything relevant is symmetrically entailed and therefore absurd. If fire were

to exist entirely separately as untouched ground from fuel (or vice versa), neither could operate on each other. In other words: If grounding is symmetrically accessible, direction collapses. If it's entirely separate from modality, it lacks modal force. If the dilemma is bridged by a further principle, the principle deals with the same problem.

5 Thus dispelling the long-held disputation between the methods of Prāsaṅgika – pure reductio, as seen in any single iteration of the formula – and Svātantrika – positive syllogisms, as expressed when one steps back and observes the broader operative structure. But that's for the next blog post.